Deep Spatial Feedback Refined Network With Multilevel Feature Fusion for Hyperspectral Image Subpixel Mapping

Junfei Zhong, Ke Wu, and Ying Xu

***Abstract*— The issue of mixed pixels is a prevalent challenge in hyperspectral images (HSIs), largely due to imaging modalities and hardware limitations. Subpixel mapping (SPM) can address this issue by distinguishing between land cover classes at a subpixel scale. In recent years, deep convolutional neural networks have demonstrated their potential and effectiveness for SPM. However, during the SPM process, the unique strengths of features at different network levels are often neglected, resulting in insufficient interaction between features at varying levels. Consequently, the network cannot fully extract and utilize the information from higher and lower level features nor can it thoroughly understand and recognize the input images. This ultimately hampers the model's SPM performance. In this study, we propose a deep spatial feedback refined network with multilevel feature fusion (SFRNet-MLFF) for SPM of hyperspectral remote sensing images. This innovative network incorporates a feedback mechanism and a multilevel feature fusion (MLFF) technique, both of which significantly enhance the interaction and aggregation of information across diverse features. Furthermore, throughout the entire network process, we apply solution space constraints to various spatial resolution contexts. These strategic implementations are expected to substantiate the network's performance in SPM. The experimental results on three hyperspectral datasets demonstrate that SFRNet-MLFF produces results with more distinct feature outlines. These results are better equipped to indicate the precise location and distribution patterns of land cover classes in low-resolution images at higher spatial resolutions. Furthermore, SFRNet-MLFF significantly improves the overall accuracy (OA) compared to the state-of-the-art deep learning SPM networks (DLSPMNets).**

***Index Terms*— Deep learning, feedback mechanism, hyperspectral image (HSI), multilevel feature fusion (MLFF), subpixel mapping (SPM).**

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NOMENCLATURE

X, Y	Input low-resolution image and the subpixel mapping result.
x_{ij}	Mixed pixel located at row i and column j within X.
A, S and D	Abundance map, the mixed spectrum vector, and the downsampling matrix.
α_{ij}, $\alpha_{ij,q}$	Abundance vector of x_{ij} and the abundance value of the qth category in x_{ij}.
$\mathbb{S}$	Predefined subpixel mapping scale.
m_{qs}	Value of the sth subpixel within x_{ij}.
SV_{qs}	Soft value of qth category for the sth subpixel.
P	One-hot processed high-resolution classification map.
$\hat{P}$	Predicted high-resolution probability map.
p_{ij}, $p_{ij,q}$	Subpixel probability vector and the value of the qth category in it.
$f_{\mathrm{MRSIM}}(\cdot)$	Multiresolution spatial information modulation module.
$f_{\mathrm{Involution}}(\cdot)$	Involution operator.
$f_{\mathrm{Downsample}}(\cdot)$	Downsampling operation.
$\Psi_{i,j}$	Neighborhood corresponding to position (i, j) on the feature map.
$H_{i,j}$	Involution kernel.
ϕ	Generation function of the involution kernel.
$f_{\mathrm{AEFE}}(\cdot)$	Attention-embedded feature enhancement function.
$f_{\mathrm{CAFF}}(\cdot)$	Cross-attention-based feature fusion function.
σ	Sigmoid function.
MaxPool$(\cdot)$	Maximum pooling operation.
AvgPool$(\cdot)$	Average pooling operation.
FC	Fully connected layer.
Max$(\cdot)$	Operation of obtaining the maximum value.
Mean$(\cdot)$	Operation of calculating the average value.

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$[\cdot]$	Features concatenation operation.
Q, K, V	Matrices of query, key, and value.
$f_{\text{Linear}}(\cdot)$	Linear transformation function.

I. Introduction

HYPERSPECTRAL imaging technology captures both the geometric spatial information and extensive spectral data of the targets, resulting in hyperspectral images (HSIs) with continuous and narrow spectral bands. This wealth of spectral information enhances the image's capacity to differentiate between various targets. Consequently, HSIs find extensive applications in diverse fields such as precise land cover classification [1], [2], [3], environmental pollution monitoring [4], [5], and rock and mineral identification [6], [7]. However, due to limitations related to the sensor's hardware and environmental interference, remote sensing images with high spectral resolution often struggle to achieve high spatial resolution [8]. This leads to most pixels in a hyperspectral remote sensing image typically corresponding to the mixed reflectance spectra of multiple pure land cover classes, thereby giving rise to the mixed pixel problem [9]. Simultaneously, each pixel in an HSI records information from dozens or even hundreds of consecutive wavebands in the instantaneous field-of-view. This results in a strong correlation between neighboring wavebands, leading to an abundance of redundant spectral information. This redundancy can adversely affect the determination of the distribution of land cover classes in the image [1], [10].

The spectral unmixing (SU) method has been proposed to address the mixed pixel problem [11], [12], [13], [14], [15], [16], [17], [18]. This method extracts pure pixels, commonly referred to as endmembers, to identify various land cover classes and determine their respective occupancy within each mixed pixel. Subsequently, it generates an abundance map for each land cover class [13]. Unlike hard classification (HC) methods, which assign a single land cover category to each mixed pixel in the remote sensing image, SU employs a soft classification approach to alleviate the mixed pixel problem to a certain extent. However, the abundance maps produced by SU still retain the original spatial resolution. They are unable to depict the fine spatial distribution of each land cover class within the mixed pixels. Consequently, the mixed pixel problem's negative impact on determining the spatial distribution of land cover classes remains unresolved [8]. To investigate the potential fine spatial distribution patterns of land cover classes at a higher spatial resolution, the concept of subpixel mapping (SPM) has been presented [19] based on the abundance maps generated by SU. In this methodology, each mixed pixel is subdivided into smaller pixel units, known as subpixels, in accordance with a predetermined scale. Each subpixel is then assigned a land cover class label, facilitating the generation of classification maps with higher spatial resolution. This process, referred to as unmixing-then-mapping (UTM), can to some extent mitigate the uncertainty in the spatial distribution of land cover classes that arises due to the mixed pixel problem [8].

Nevertheless, the process of mapping from low resolution (LR) to high resolution (HR) is an ill-posed problem [20], characterized by a lack of spatial clues regarding the true geographic distribution within mixed pixels. These clues are lost during the image imaging process, leading to a diversity of final results. This necessitates the introduction of relevant spatial feature prior information to constrain the solution space of the SPM. Spatial feature prior information can be broadly categorized into two types based on its sources and characteristics: algorithm-based fixed priors and external dataset-based optional priors. The former leverages the data to be processed itself to identify the most suitable SPM results, based on reasonable relationships between pixels or subpixels. Classical methods in this category include SPM based on spatial attraction [21], [22], pixel-swapping algorithm (PSA) [23], [24], maximum a posteriori (MAP) and spatial attraction [25], interpolation [26], deconvolution [27], and spectral–spatial correlation [28], [29], among others. On the other hand, external dataset-based optional priors combine relevant information from external datasets to constrain the SPM results of the target data. Classical methods in this category include backpropagation-based SPM [30] and neural network-based SPM [31] that rely on external LR and HR image pairs, SPM based on the complementarity of multitemporal images with different resolution using various spatial-temporal fusion methods [32], [33], [34], and so on. However, algorithm-based fixed priors are often inflexible and overly simplistic, rendering them unsuitable for surface areas with high spatial heterogeneity, particularly areas with highly mixed distributions of multiple land cover classes [8], [20]. Optional priors based on external data typically lack generalizability and are only suitable for local area cases [20], and the availability of external datasets cannot always be guaranteed. As such, there is a need for a more stable and reliable methodology.

In recent years, the field of SPM has greatly benefited from the robust learning and feature representation capabilities of deep learning, leading to the emergence of a variety of deep learning-based SPM networks (DLSPMNets). These DLSPMNets leverage the most suitable complex mapping functions, learned from training data of the same source or with similar characteristics, as priors for the SPM process of the target data. This approach effectively overcomes the limitations of traditional priors and provides comparatively more accurate constraints on the solution space, demonstrating remarkable capability and impressive results in recovering and presenting high-resolution land cover details. Existing DLSPMNets can be generally categorized into two trends. The first trend aligns with the philosophy of classical SPM approaches, relying on the abundance maps generated by SU at an LR to establish a mapping between them and the high-resolution classification maps using the networks. Ling and Foody [35] proposed a deep learning-based super-resolution mapping algorithm (DeepSRM) [20], which utilized LR abundance maps as inputs to the network to estimate HR indicator maps for each category. Ma et al. [36] put forward a super-resolution convolutional neural network-based SPM method (SRCNN-SPM), which employs a transfer learning strategy to restore the LR abundance maps to the HR space. Zhang et al. [37] proposed a super-resolution mapping (SRM) method based on graph convolutional network (SRM_{GCN}),

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aiming to enhance SRM results by capturing structural information on the graph. The significance of UTM methods based on LR abundance maps cannot be understated. However, these methods present a notable challenge—errors in the abundance maps are propagated to the SPM results, leading to an accumulation of errors. This is particularly evident in areas with complex surface distributions. In contrast, the other trend in DLSPMNets integrates SU and SPM processes, establishing end-to-end networks to derive HR classification (HRC) results using LR images as input. Jia et al. [38] proposed an SRM network (SRM_{CNN}) based on convolutional neural networks to model the nonlinear relationship between LR remote sensing images and HR land cover maps. He et al. [8] proposed a generic framework for designing DLSPMNet, which led to the construction of a semantic information modulation network (SIMNet) for SPM [20], using LR classification maps as the semantic prior to enhance spatial contextual feature representation. Zhang et al. [39] proposed a soft information-constrained network (SCNet) that extracted multilevel feature representations from both remote sensing images and prior soft information, integrating these features into image features. In terms of applications, Wang et al. [40] designed a novel spatial–temporal depth-wise residual network (ST-DRes) model for SPM of different crop types from moderate-resolution imaging spectroradiometer (MODIS) images, efficiently and effectively addressing spatial and temporal data. He et al. [41] proposed context-aware subpixel mapping network (CASNet) to generate a 2-m fine-scale urban tree cover product from 10-m Sentinel-2 images over 34 metropolises in China. Jia et al. [42] proposed a spatial relationship-informed super-resolution mapping network (SRSNet) to extract 2.5-m rural road maps from Sentinel-2 images. He et al. [43] developed a learnable correlation-based SPM (LECOS) method to model teleconnections that facilitate correlations between remote locations or objects and to adaptively learn the varied global correlation patterns between pixel and subpixel, thereby enabling the inference of subpixel-scale distribution for each land cover class to indicate urban pattern dynamics. This integrated modeling approach allows the network to learn features from the original images and avoids the cumulative errors of step-by-step modeling, generating better results. This integrated modeling approach allows the network to learn features from the original images, circumventing the cumulative errors of step-by-step modeling and yielding superior results. This end-to-end DLSPMNets trend is currently a primary research focus in the SPM field and is also the central theme of this article.

Current end-to-end DLSPMNets predominantly utilize convolutional neural networks. In these networks, the shallow layers extract low-level features that focus more on detailed information, encompassing more fine-grained features such as spatial geometric features, e.g., contours and shapes. As the network deepens, these low-level features are transformed into high-level features that pay more attention to semantic information, providing a more abstract and generalized description of the image content. The spatial detail characteristics inherent in the low-level features are vital to the SPM task. However, these characteristics gradually become blurred as they propagate forward in the network. In addition, the abstract and comprehensive feature representation of high-level features can better capture the essential features of the images, revealing more complex concepts and relationships within the images, which also significantly influences the SPM effect. The differences between these different levels of features are often overlooked, making it necessary to investigate the impact of utilizing multilevel features on SPM performance. Simultaneously, the solution space of SPM needs to be effectively constrained to enhance the reliability of SPM results. Furthermore, different land cover categories in HSIs exhibit distinct spatial and spectral distribution characteristics and correlations. This necessitates the network to possess a more flexible discriminative capability toward the network features. Consequently, the learning of spatial and spectral weights in the feature maps becomes essential for DLSPMNets. This process aids in enhancing the representation of spatial and spectral features, thereby improving the performance of SPM.

To facilitate interaction between different levels of features, a feedback mechanism is introduced. This mechanism incorporates a spatial information modulation function to preserve the spatial detail information inherent in the original images and constrain the SPM solution space at various resolutions. In addition, a multilevel feature fusion (MLFF) module is designed to enhance spectral and spatial correlations and to fuse information from multilevel features. Building on this foundation, we propose a deep spatial feedback refined network with MLFF (SFRNet-MLFF) for HSI SPM. The primary contributions of this article can be summarized as follows.

1) SFRNet-MLFF is proposed for HSI SPM. Initially, the input image is transformed into multilevel network features via a multiple-unfolding feedback structure. This process also incorporates spatial information modulation at varying spatial resolutions. These multilevel features are subsequently processed through an MLFF module, which performs both feature enhancement and fusion processes. Ultimately, the fused features are converted into the network output by the classifier following an upsampling and global spatial information modulation (GSIM) process.
2) The feedback refined structure incorporated in the SFRNet-MLFF ensures the generation of more robust higher level features by adjusting LR lower level features with medium-resolution (MR) feedback features. Simultaneously, the modulation of spatial information at different spatial resolutions aids in preserving and recovering spatial details inherent in the original images, and to a certain extent, constraining the SPM solution space.
3) To enhance the spectral and spatial correlation of each land cover class, spectral and spatial attention (SA) mechanisms are employed to facilitate the adaptive information weights learning of network features. Subsequently, the feature fusion based on the cross-attention mechanism enables the aggregation and communication of multilevel features, leveraging the details of the low-level features and the semantics of the high-level features.

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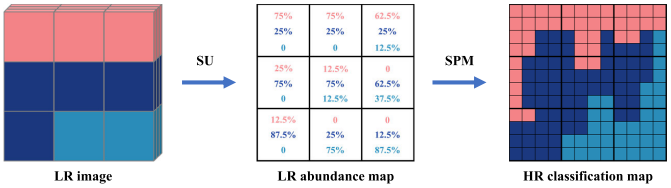


Fig. 1. Process of the UTM SPM method.

4) Experiments were conducted on three publicly available hyperspectral datasets to compare the proposed method with classical SPM methods and the state-of-the-art DLSPMNets. The results demonstrate that the proposed SFRNet-MLFF generates SPM results on HSIs with clearer detail features and surface contours, which more accurately represent the real geographic distribution patterns of each land cover class within the mixed pixels. Moreover, SFRNet-MLFF exhibits superior quantitative evaluation performance, significantly outperforming other methods.

II. Background

A. UTM SPM Formulation

SPM enables the identification of the geographic distribution of each land cover category within mixed pixels at a subpixel level, based on a predefined scale. UTM SPM methods are grounded on the abundance maps produced by the SU process on LR images. Fig. 1 provides an illustration of the UTM process. In this process, an area comprising 3×3 mixed pixels undergoes the SU operation to yield an abundance map of the three categories. This map indicates the proportion of each category within each mixed pixel. Subsequently, each mixed pixel is divided into subpixels according to a predefined scale (set to 4 in the figure, implying that each mixed pixel is subdivided into 16 subpixels). The category label to which each subpixel belongs is then assigned to generate the HRC result.

The mathematical derivation of UTM SPM commences with the SU based on a provided LR image

$$X = \begin{bmatrix} [[x_{11}, x_{12}, \ldots, x_{1w}], [x_{21}, \ldots, x_{2w}], \ldots, [x_{h1}, \ldots, x_{hw}]]_1 \\ [[x_{11}, x_{12}, \ldots, x_{1w}], [x_{21}, \ldots, x_{2w}], \ldots, [x_{h1}, \ldots, x_{hw}]]_2 \\ \cdots \\ [[x_{11}, x_{12}, \ldots, x_{1w}], [x_{21}, \ldots, x_{2w}], \ldots, [x_{h1}, \ldots, x_{hw}]]_c \end{bmatrix}.$$

In this context, $X \in \mathbb{R}^{h \times w \times c}$, where h, w, and c denote the value of height, width, and band, respectively. Each of these mixed pixels $x_{ij} \in \mathbb{R}^{(1\times1)\times c} (i \in h, j \in w)$ can be collectively represented by the abundance vector $\alpha_{ij} \in \mathbb{R}^{(1\times1)\times NC} (i \in h, j \in w)$ and the mixed spectrum vector $S \in \mathbb{R}^{\mathrm{NC}\times c}$ [44]. Here, NC signifies the number of categories

$$x_{ij} = \alpha_{ij} S + b. \tag{1}$$

Thus, the LR image $X \in \mathbb{R}^{(h\times w)\times c}$ can be represented by the corresponding LR abundance map $A \in \mathbb{R}^{(h\times w)\times \mathrm{NC}}$ and the mixed spectrum vector S

$$X = AS + b$$
$$A = \begin{bmatrix} [\alpha_{11}, \alpha_{12}, \ldots, \alpha_{1w}] \\ [\alpha_{21}, \alpha_{22}, \ldots, \alpha_{2w}] \\ \cdots \\ [\alpha_{h1}, \alpha_{h2}, \ldots, \alpha_{hw}] \end{bmatrix}, \text{ s.t. } \alpha_{ij,q} \geq 0, \ \sum_{q=1}^{\mathrm{NC}} \alpha_{ij,q} = 1 \tag{2}$$

where $\alpha_{ij,q}$ is the abundance value of the qth category within the mixed pixel x_{ij}. Subsequently, each mixed pixel is subdivided into smaller subpixels according to a predefined SPM scale $\mathbb{S}$. The objective function is then maximized based on the soft values (such as spatial attraction values [21]) of the subpixels under the constraint of the abundance map A. This process yields the final SPM result Y [8], [36]

$$Y = \text{argmax} \left(\sum_{k=1}^{\mathrm{NC}} \sum_{s=1}^{\mathbb{S}^2} m_{qs} \times \mathrm{SV}_{qs} \right)$$
$$\text{s.t.} \sum_{q=1}^{\mathrm{NC}} m_{qs} = 1, \sum_{s=1}^{\mathbb{S}^2} m_{qs} = \alpha_{ij,q} \times \mathbb{S}^2 \tag{3}$$

where m_{qs} is the value of the sth subpixel within the mixed pixel x_{ij}. It assumes a value of 1 when the sth subpixel is assigned to the qth category, and otherwise, it is 0. SV_{qs} denotes the soft value of qth land cover category for the sth subpixel.

B. End-to-End DLSPMNet Formulation

The deep learning networks that establish the mapping relationship between LR images and HRC results have a solid mathematical foundation. This end-to-end mapping, which integrates SU and SPM, avoids errors that may occur during the UTM process. Moreover, it allows the models to learn and utilize the original information from the LR images more comprehensively and thoroughly. Generally, an end-to-end DLSPMNet can be divided into three parts based on their functions [8], [20], as depicted in Fig. 2.

1) The feature extraction and transformation part, denoted as $f_F(\cdot)$, is used to extract features from LR images and transform these features within the LR feature space.
2) The upsampling part, represented as $f_U(\cdot)$, is responsible for mapping LR features into the HR space.
3) The classification part, $f_C(\cdot)$, is designed to map HR feature maps into probability maps for each respective category.

Given a training patch pair that consists of an LR image X and a one-hot processed HRC map P, the end-to-end DLSPMNet, which includes the aforementioned three parts, initially conducts a function transformation on the LR image X to get the predicted HR probability map $\hat{P} \in \mathbb{R}^{(H\times W)\times(1\times \mathrm{NC})}$

$$\hat{P} = f_C(f_U(f_F(X))). \tag{4}$$

Subsequently, the network parameters are iteratively updated in alignment with the target P, aiming to converge the network by minimizing the loss function $L(\hat{P}, P)$. Ultimately, the thoroughly trained network is employed to generate new results corresponding to the new LR images.

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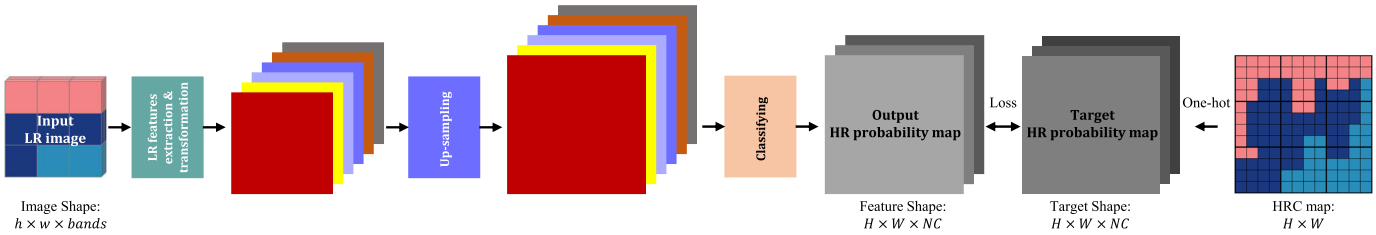


Fig. 2. Process of the end-to-end DLSPMNet.

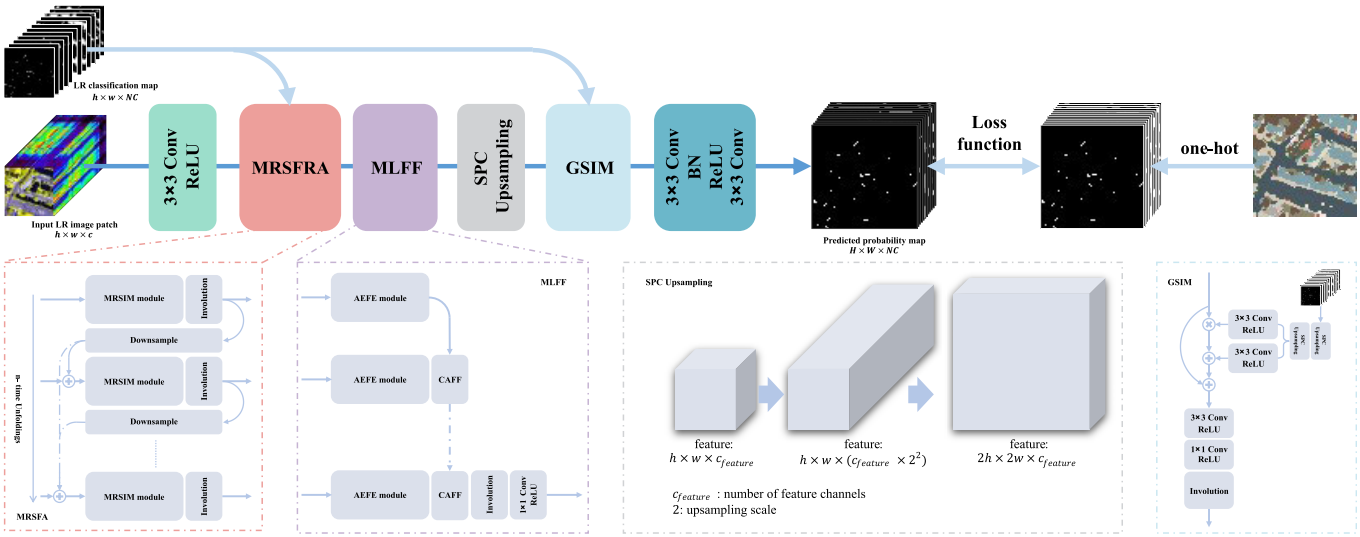


Fig. 3. Structure of SFRNet-MLFF.

For the mathematical derivation of this end-to-end mapping, the initial step also commences with SU, the processes and outcomes of which are referred to in (1) and (2). Following this, the LR abundance map A and the HR abundance map (also known as HR probability map) $P \in \mathbb{R}^{(H\times W)\times NC}$ for the NC categories can be interconnected by the degradation matrix $D \in \mathbb{R}^{(h\times w)\times(H\times W)}$ [20]:

$$A = DP. \tag{5}$$

Drawing from (2) and (5), a functional mapping relationship is established between the LR image X and the HR abundance map P. This relationship is crucial for the integrated modeling of SU and SPM

$$\begin{aligned} &X = DPS + b \\ &P = \begin{bmatrix} [p_{11}, p_{12}, \ldots, p_{1W}] \\ [p_{21}, p_{22}, \ldots, p_{2W}] \\ \cdots \\ [p_{H1}, p_{H2}, \ldots, p_{HW}] \end{bmatrix}, \quad p_{ij} \in \mathbb{R}^{(1\times 1)\times \mathrm{NC}}, \ i \in H, j \in W \\ &\text{s.t. } p_{ij,q} \geq 0, \sum_{q=1}^{\mathrm{NC}} p_{ij,q} = 1. \end{aligned} \tag{6}$$

The final form of (6) is derived by designating the LR image X as input and the HR abundance map P as output

$$P = D^{-1}(X - b)S^{-1}. \tag{7}$$

The objective of the deep learning network model, grounded on this mathematical foundation, is to obtain the optimal solution achieved by training the network to implicitly learn the parameters D^{-1}, S^{-1}, and b. In essence, this implies that the intricate functional relationship between the LR image and the HRC result is approximated through the extensive parameters of the network.

III. Proposed SFRNet-MLFF Algorithm

To foster the interplay of features at various levels and bolster the precision and robustness of feature representations, we introduced a feedback mechanism. This mechanism incorporates spatial information modulation processes, seamlessly integrating LR classification maps into the network structure. The feature fusion module was designed within the network to enhance features through adaptive weight learning and to facilitate MLFF. The comprehensive structure of the proposed SFRNet-MLFF is depicted in Fig. 3. In SFRNet-MLFF, the input LR HSI patches are sequentially transformed into HR features via a convolution block, which includes a convolution layer and a rectified linear unit (ReLU) function, multiresolution spatial feedback refined architecture (MRSFRA), MLFF, and a subpixel convolution (SPC) [45]. Subsequently, these HR features are mapped into the HR probability maps of each category through the GSIM module and a classifier composed of a batch normalization (BN) layer, a ReLU function, and two convolutional layers. The cross-entropy loss between the SFRNet-MLFF output and ground truth (GT) was selected as the model loss function. The model's loss function is the cross-entropy loss between the SFRNet-MLFF output

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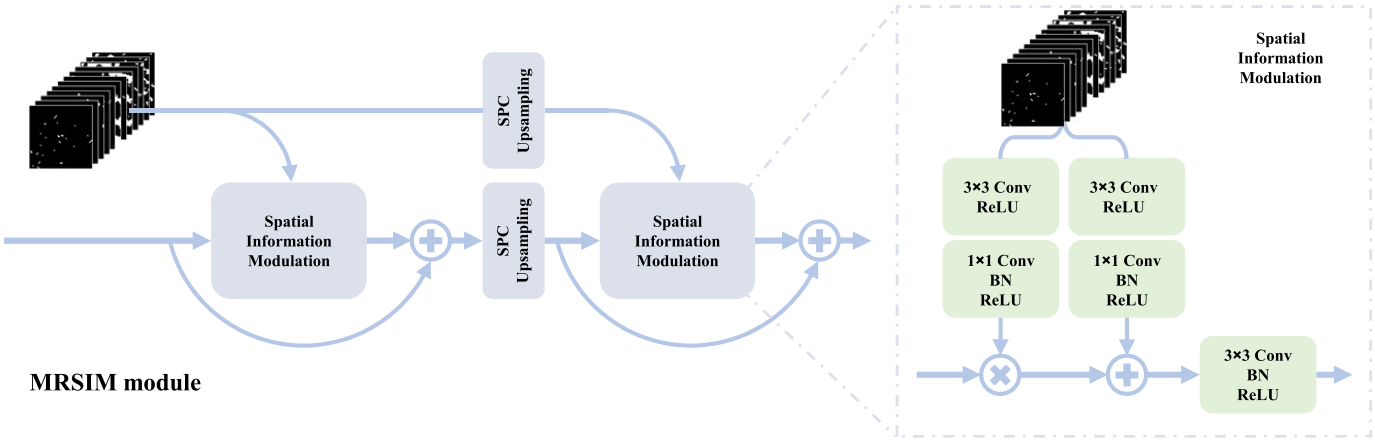


Fig. 4. Structure of MRSIM module.

and the GT. The model parameters are updated during the backpropagation of the loss function gradient.

The proposed SFRNet-MLFF comprises three primary parts.

1) The MRSFRA incorporates a feedback mechanism that facilitates the adjustment to lower level features, thereby generating more robust higher level features. Simultaneously, multiple unfoldings of the feedback structure yield multilevel output features that encompass multilevel properties. Furthermore, a multiresolution spatial information modulation (MRSIM) module is integrated to ensure the preservation of the original images' spatial features and to constrain the SPM solution space at two different resolution contexts.
2) The MLFF module conducts feature enhancement and MLFF on the multilevel output features derived from the preceding stage. This process fully leverages multilevel features, amalgamates detail and semantic information, and yields superior results.
3) The GSIM module is adept at adjusting the network features at the target spatial resolution. This creates a continuous-resolution-context enhancement on spatial information in conjunction with MRSFRA and provides a robust global constraint on the SPM results' solution space.

A. Multiresolution Spatial Feedback Refined Architecture

Fig. 3 further showcases our proposed MRSFRA. This architecture incorporates an MRSIM module, an involution operation, and a downsampling process. It is characterized by its n-time unfoldings, where n is set to 4 in the experiments. A key characteristic of MRSFRA is its capacity to convey higher level information from MR layers to the preceding layers. This transmission process refines the expression of LR lower level information via feedback connections, in turn generating a more robust higher level expression [46]. As the architecture unfolds, it generates multilevel network features, which are then fed into the subsequent module.

In MRSFRA, the output feature $\boldsymbol{F}^k_{\mathrm{MRSFRA_{Out}}}$ for the kth unfolding is derived after the input feature $\boldsymbol{F}^k_{\mathrm{MRSFRA_{In}}}$ passes through the corresponding MRSIM module f^k_{MRSIM} and the involution operator $f^k_{\mathrm{Involution}}$. This process can be represented as

$$\boldsymbol{F}^k_{\mathrm{MRSFRA_{Out}}} = f^k_{\mathrm{Involution}}\left(f^k_{\mathrm{MRSIM}}\left(\boldsymbol{F}^k_{\mathrm{MRSFRA_{In}}}\right)\right) \tag{8}$$

and the $(k+1)$th input $\boldsymbol{F}^{k+1}_{\mathrm{MRSFRA_In}}$ can be expressed as

$$\boldsymbol{F}^{k+1}_{\mathrm{MRSFRA_{In}}} = \boldsymbol{F}^k_{\mathrm{MRSFRA_{In}}} + f^k_{\mathrm{Downsample}}\left(\boldsymbol{F}^k_{\mathrm{MRSFRA_{Out}}}\right) \tag{9}$$

where $f^k_{\mathrm{Downsample}}$ denotes the kth downsampling process.

1) MRSIM Module: Fig. 4 illustrates our design of the MRSIM module. Unlike conventional methods that limit the restoration and enhancement of spatial details to the LR feature space, the MRSIM module is designed to incorporate the representation of network features at higher spatial resolution. This approach facilitates the recovery of multiresolution spatial features and constrains the SPM solution space. In the MRSIM module, the feature maps are initially constrained by the spatial information of the LR classification maps for each category and modulated at the original spatial resolution. They are then converted to MR after SPC [45] upsampling (as illustrated in Fig. 3) and are further constrained by the classification maps of the same resolution (SPC-operated LR classification maps). Simultaneously, the input features for each modulation are directly passed to the output features via a skip connection, preventing network degradation issues. This design of MRSIM module effectively complements spatial features and restricts the solution space in continuous spatial resolution contexts, thereby enhancing the performance of SPM.

During each spatial information modulation, the classification information is processed through two identical convolutional structures. This processing generates a set of multiplicative and incremental quantities, which are subsequently integrated directly into the network's main framework, as elaborately illustrated in Fig. 4.

2) Involution and Downsample: The involution operator [47], by inverting the design principles of convolution—being spatial-agnostic and channel-specific, exhibits properties that are spatial-specific and channel-agnostic. This allows it to aggregate contextual information across a broad spatial range while adaptively assigning weights at different spatial locations, all the while sharing meta-weights.

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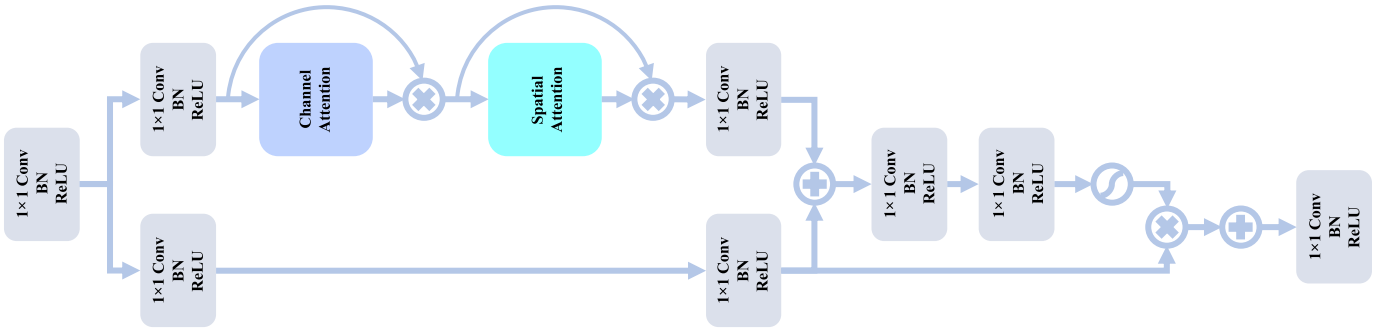


Fig. 5. Structure of AEFE.

Specifically, for each position (i, j) on the feature map $\boldsymbol{F}$, a corresponding neighborhood $\Psi_{i,j}$ is identified. The kernel generation function ϕ is then used to generate the corresponding involution kernel $H_{i,j}$ for that position

$$H_{i,j} = \phi\left(\Psi_{i,j}\right). \tag{10}$$

Within the network, the process can be expressed as follows:

$$H_{i,j} = W_1\left(\mathrm{ReLU}\left(\mathrm{BN}\left(W_0\Psi_{i,j}\right)\right)\right) \tag{11}$$

where W_0 and W_1 denote the convolution layers.

Ultimately, this involution kernel is used to execute computational operations on the neighborhood $\Psi_{i,j}$ in the original feature map $\boldsymbol{F}$. In MRSFRA, we have incorporated the involution module subsequent to each MRSIM module to aggregate spatial context information and initially allocate weights to various spatial locations of network features. Following this, we convey the higher level features, which have been upscaled by the SPC upsampling in the MRSIM module, to the LR space of the original input via a downsample process in one output branch. This corresponds to the input of the subsequent unfolding of the MRSFRA.

B. Multilevel Feature Fusion

In the network, low-level features with smaller receptive fields preserve more detailed information, while high-level features with larger receptive fields contribute semantic knowledge pertinent to image region categories. To leverage these multilevel features effectively, we have devised the MLFF module, as depicted in Fig. 3. This module fuses the final features from each forward output of MRSFRA, amalgamating detailed and semantic information. The process primarily encompasses two stages: attention-embedded feature enhancement (AEFE) and cross-attention-based feature fusion (CAFF). The process can be expressed as follows:

$$\boldsymbol{F}^k_{\mathrm{AEFE_{Out}}} = f^k_{\mathrm{AEFE}}\left(\boldsymbol{F}^k_{\mathrm{MRSFRA_{Out}}}\right) \tag{12}$$

$$\boldsymbol{F}^v_{\mathrm{CAFF_{Out}}} = f^v_{\mathrm{CAFF}}\left(\boldsymbol{F}^k_{\mathrm{AEFE_{Out}}}, \boldsymbol{F}^{k+1}_{\mathrm{AEFE_{Out}}}\right) \tag{13}$$

where $\boldsymbol{F}^k_{\mathrm{MRSFRA_{Out}}}$ represents the output feature of the kth unfolding of MRSFRA; f^k_{AEFE} denotes the corresponding kth feature enhancement process; $\boldsymbol{F}^k_{\mathrm{AEFE_{Out}}}$ and $\boldsymbol{F}^{k+1}_{\mathrm{AEFE_{Out}}}$ are the output features of the kth and the $(k+1)$th feature enhancement respectively; f^v_{CAFF} is the feature fusion process corresponding to these two feature enhancements; and $\boldsymbol{F}^v_{\mathrm{CAFF_{Out}}}$ is the output feature of this feature fusion process.

1) Attention-Embedded Feature Enhancement: The AEFE module, which incorporates channel attention (CA) and SA mechanisms [48], is designed to enhance the input features, as depicted in Fig. 5. The overall architecture of this module is inspired by the residual attention module (RAM) [49]. This design strategy allows for the preservation of the original information contained within the input features while simultaneously bolstering the correlation expression of the spectral and spatial features [10], ultimately facilitating the achievement of feature enhancement. CA and SA are embedded to sequentially generate attention maps from the network's feature maps, operating along two distinct dimensions—channel and spatial. The processes are illustrated in Fig. 6.

The CA mechanism adaptively learns the weights of feature map channels by concentrating on the interdependencies among the channels. It generates a channel-wise attention map that is then used to adjust the original feature map [48]. This method enables the CA mechanism to adaptively discern which channels are more crucial for the given task, thereby enhancing the representational capacity of the feature maps. The computational process of the CA mechanism for the feature map $\boldsymbol{F}$ is expressed as follows:

$$\boldsymbol{F}_{\mathrm{CA_{out}}} = \sigma\left(\mathrm{FC}(\mathrm{MaxPool}(\boldsymbol{F}))+\mathrm{FC}(\mathrm{AvgPool}(\boldsymbol{F}))\right) \tag{14}$$

where MaxPool$(\cdot)$ refers to the maximum pooling operation, AvgPool$(\cdot)$ denotes the average pooling operation, and FC stands for the fully connected layer. σ represents the sigmoid function, and $\boldsymbol{F}_{\mathrm{CA_{out}}}$ is the output feature of the CA mechanism.

The SA mechanism, akin to the CA mechanism, adaptively learns the weights of feature maps by focusing on the interdependencies among spatial locations in a feature map. It generates an SA map to reweight the original feature map [48]. This enables the SA mechanism to adaptively identify which spatial locations are more important for the task, thereby improving the feature representations. The computation process of SA mechanism for the feature map $\boldsymbol{F}$ is expressed as follows:

$$\boldsymbol{F}_{\mathrm{SA_{out}}} = \sigma\left(W[\mathrm{Max}(\boldsymbol{F}), \mathrm{Mean}(\boldsymbol{F})]\right) \tag{15}$$

where Max$(\cdot)$ denotes the operation of obtaining the maximum value, Mean$(\cdot)$ signifies the operation of calculating the average value, and $[\cdot]$ represents the concatenation operation. W stands for the convolutional layer and σ is the sigmoid function. $\boldsymbol{F}_{\mathrm{SA_{out}}}$ is the output feature of the SA.

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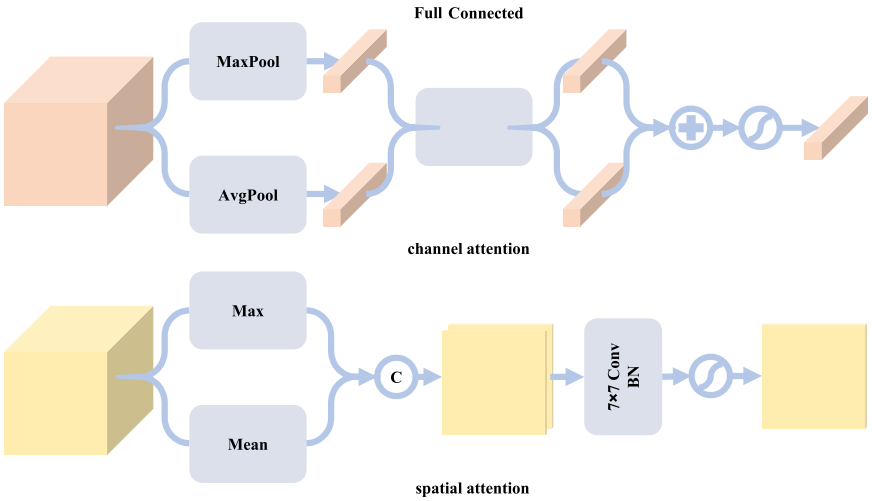


Fig. 6. Channel attention and spatial attention.

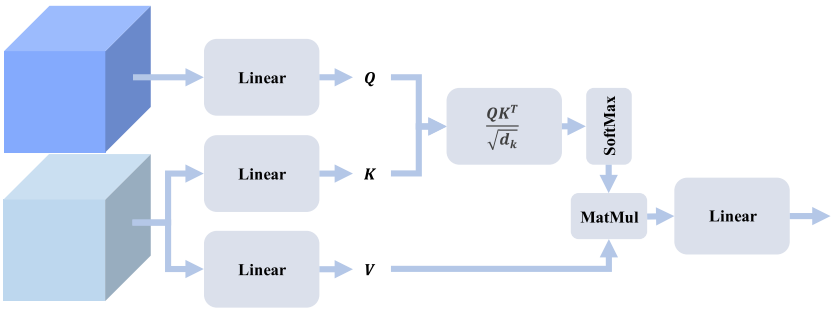


Fig. 7. Cross-attention mechanism.

2) Cross-Attention-Based Feature Fusion: Feature fusion, a widely utilized technique, serves to bolster information aggregation within the network [50], [51], [52]. Common strategies for feature fusion encompass point-wise addition (PWA) [53], [54], [55] and the direct concatenation of feature maps. However, these methods lack focus on multilevel features and can directly impact the fusion effect. To address this, we have implemented feature fusion based on the cross-attention mechanism [56] (CAFF). This approach integrates different levels of information and establishes connections between various feature maps, facilitating interaction and fusion of multilevel features. As a result, it generates comprehensive features that encompass both spatial information and rich semantics, thereby enhancing the performance of the model.

As depicted in Fig. 7, within the cross-attention mechanism, two distinct features, $\boldsymbol{F}_1$ and $\boldsymbol{F}_2$, undergo linear transformations to yield three different feature variables: Q, K, and V. These variables are subsequently processed to derive the fused output feature, $\boldsymbol{F}_{\text{out}}$. The procedure can be expressed as follows:

$$\begin{aligned} Q &= f_{\text{Linear}_1}(\boldsymbol{F}_1) \\ K &= f_{\text{Linear}_2}(\boldsymbol{F}_2) \\ V &= f_{\text{Linear}_3}(\boldsymbol{F}_2) \end{aligned} \tag{16}$$

$$\boldsymbol{F}_{\text{out}} = f_{\text{Linear}_4}\left(\text{SoftMax}\left(\frac{QK^T}{\sqrt{d}_K}\right)V\right) \tag{17}$$

where $f_{\text{Linear}_i}, i = 1, 2, 3, 4$, represents the linear transformations. SoftMax refers to the SoftMax function, and $(d)_K^{1/2}$ is the scaling factor, which is determined by the dimension number of K.

C. GSIM Module

Following the feature fusion process, the output network features are once more upsampled to the HR space via the SPC layer. To supplement the spatial details and limit the SPM solution space within the HR context, we introduce the GSIM module. In this module, the classification information extracted from the upsampled LR classification maps is adaptively transformed into a series of multiplications and increments that are integrated into the network. Ultimately, the network features undergo another round of processing by the involution operator before they are outputted. This structural design, as illustrated in Figs. 3 and 8, ensures the enhancement of the spatial features and the restriction of the SPM solution space at the desired spatial resolution.

IV. Experiments

Datasets: In our experiments, three publicly available hyperspectral datasets were considered: the Houston dataset, the Pavia University dataset, and the Kennedy Space Center (KSC) dataset. For these HR public datasets, acquiring perfectly registered LR images to create pairs of LR images and HRC data for network training and testing is challenging. Therefore, we downsampled the original public data images using a

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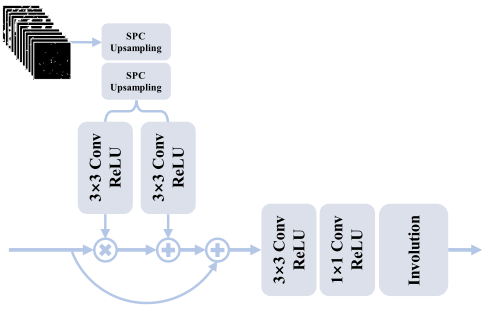


Fig. 8. Structure of GSIM.

predetermined scale (set to 4 in the experiments) to generate LR images. This approach guarantees the precise alignment between the LR and HR images.

Preparations: Given that each of the three datasets comprises a single HSI, we generated patch pairs from the downsampled LR images and the GT of the original HR images. A random proportion ρ was then applied to select the network training data, with the remaining data used for network testing. In the experiments, the GT, with the shape of $H \times W$, underwent one-hot processing to reshape it to NC $\times$ $H \times W$. The LR classification maps for each category were derived by downsampling the one-hot processed GT. For the end-to-end DLSPMNets, the LR image patch size cropped from the data is bands $\times$ 16 $\times$ 16, and the one-hot processed HR GT patch size is NC $\times$ 64 $\times$ 64. Here, bands and NC vary with different datasets, and the corresponding training sample proportion ρ differs for different datasets. For the UTM SPM methods that take abundance maps as input, we referred to and improved upon the related methods [8], [20]. We applied the ρ ratio to the endmember samples in LR images to select the endmembers for extracting endmember spectra, which were then used in SU to obtain the abundance maps of all categories. This approach allowed for a comparison with end-to-end DLSPMNets that take LR images as input. Consequently, we were able to effectively evaluate the SPM performance of various methods.

Comparison Methods: To evaluate the performance of the proposed SFRNet-MLFF, we selected five state-of-the-art end-to-end DLSPMNets for comparative experiments. These include SRM_{CNN} [38], subpixel mapping convolutional neural network-efficient subpixel convolutional neural network (SPMCNN-ESPCN) [8], CASNet [41], SCNet [39], and SIMNet [20]. In addition, we chose three UTM SPM methods, namely, spatial attraction based subpixel mapping (SASPM) [21], PSA [23], and SRCNN-SPM [36], for comparison. To further demonstrate the effectiveness of SPM in addressing mixed pixels, we used the selected endmembers as regions of interest for direct HC of LR images using the support vector machine (SVM). This allowed us to compare the visual effects of the HC method and SPM methods.

Parameter Setting: For the end-to-end DLSPMNets, the training batch size of each dataset is set to 4. A fixed learning rate of 1e-4 is used, and the Adam optimizer is selected for network optimization, with β_1 and β_2 set to 0.9 and 0.999, respectively. The value of ε is set to $1e^{-8}$. The training process is set to run for 1000 epochs, and the upsampling scale from LR to HR is set to 4. No pretrained models are available for the end-to-end DLSPMNets. For SRCNN-SPM, the model is pretrained using the Berkeley Segmentation dataset (BSD 300 dataset) following its experimental methodology [36], with the same batch size, learning rate, optimizer, and epoch settings as mentioned above. The well-trained model is then transferred to generate HR probability maps of different land cover types.

Evaluation Metrics: To conduct a quantitative assessment of the proposed method, we selected the overall accuracy (OA), Kappa coefficient (κ) of the test outcomes, and the $F1$ score for each category as our evaluation metrics.

Experimental Environment: Training and testing processes are executed in PyTorch, utilizing a server equipped with a 24G NVIDIA GeForce RTX 3090 graphics processing unit (GPU).

A. Experiment 1: Houston Dataset

Houston dataset was acquired by the NSF-funded Center for Airborne Laser Mapping (NCALM). This dataset, which covers the University of Houston campus and its surrounding urban area, was collected on June 23, 2012. It comprises 144 spectral bands in the 0.38–1.05-μm region and features an image size of 349 $\times$ 1905 pixels. The spatial resolution of the dataset is 2.5 m.

In the experiment, we initially selected a subset image of 340 $\times$ 1900 [Fig. 9(a)] from the original data to facilitate data processing. This image was then downsampled according to the experimental scheme to generate the LR image [Fig. 9(b)]. The LR image and the HR GT, which was categorized into 15 land cover types [Fig. 9(c)], were cropped into 145 pairs of LR image patches and GT patches. These patches had sizes of 144 $\times$ 16 $\times$ 16 and 1 $\times$ 64 $\times$ 64. Out of these, 104 random patch pairs were chosen for network training, while the remaining 41 patch pairs were used for testing. The GT patch was one hot processed to a shape of 15 $\times$ 64 $\times$ 64. It was then downsampled to produce an LR classification map with a shape of 15 $\times$ 16 $\times$ 16, which was then inputted into the network. A part of the cropped patch pairs, along with their corresponding original HR images, are shown in Fig. 9(d). Table I provides the count of labeled pixels in the data and the ratio ρ, which represents the proportion of labeled pixels utilized for training. For the UTM methods, the endmembers [Fig. 10(b)] of the LR image were derived in accordance with the experimental scheme. We then selected the endmembers to each category involved in the SU based on the ratio ρ (Table II), extracted the spectra of the endmembers, and generated the abundance maps [Fig. 10(c)].

Five test LR image patches were chosen to illustrate the generated HRC visual results, as depicted in Fig. 11. This qualitative comparative assessment clearly highlights the drawbacks of direct HC on LR images. Due to the influence of mixed pixels, the classification of land cover categories appears highly blurred and fragmented and significantly lacks spatial details for recognition and distinction. While SPM can address the problem of mixed pixels, the UTM-based SPM methods are impacted by the high complexity of land cover distribution. They also accumulate errors produced by the SU process, resulting in a final visual effect that deviates

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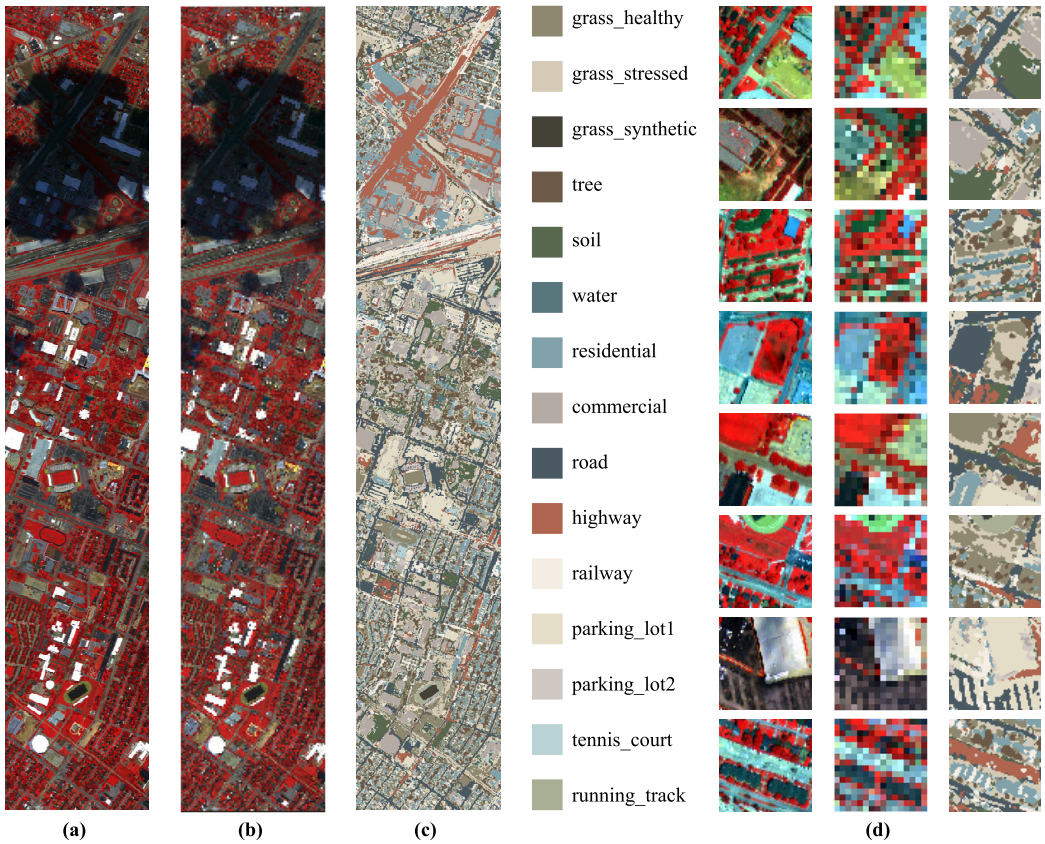


Fig. 9. Experiment on Houston dataset. (a) HR false-color image. (b) LR false-color image. (c) HR GT of the HR Houston image. (d) Part of the cropped obtained patch pairs and the corresponding original HR images.

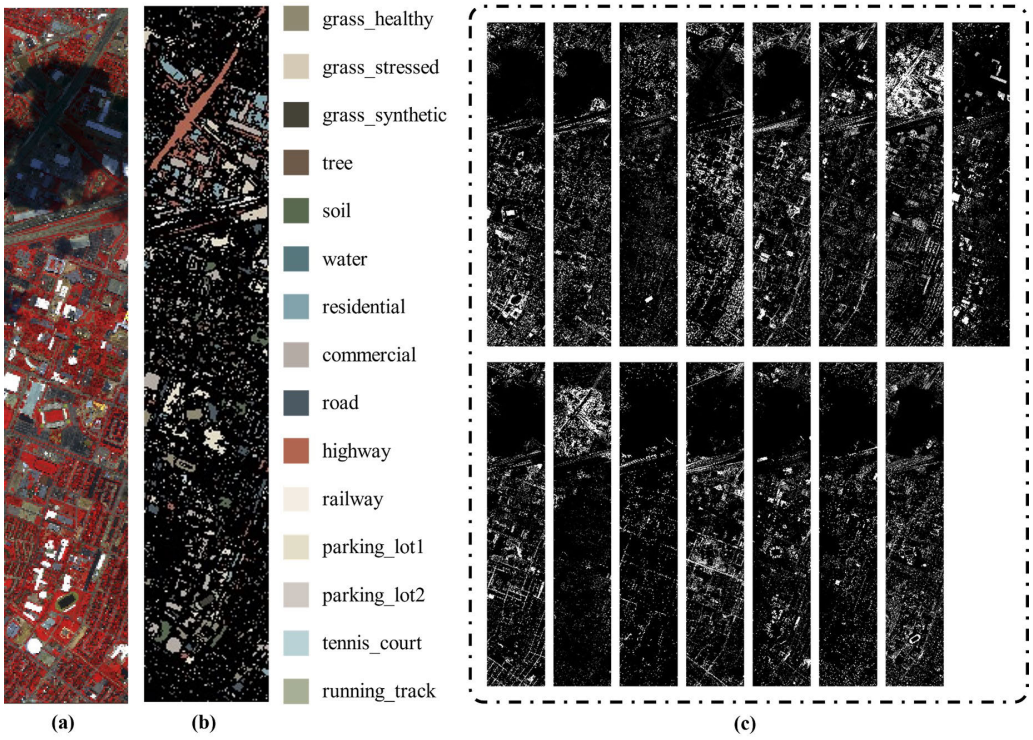


Fig. 10. Experiment on Houston dataset. (a) LR false-color image. (b) Pure pixels as endmembers. (c) Abundance maps of 15 classes.

significantly from the original GT patch used as a reference. Essentially, they fail to restore the original spatial distribution pattern and spatial details.

Comparatively, the end-to-end DLSPMNets produce more visually appealing results, with a range of methods showing varying levels of improvement over HC and UTM SPM,

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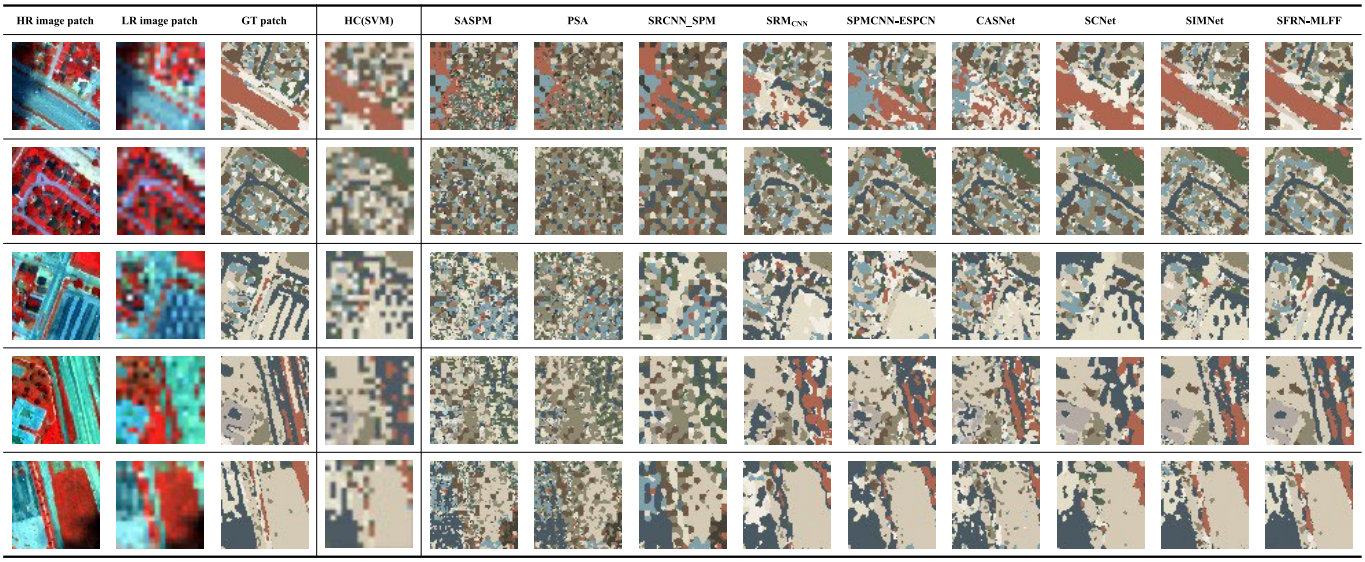


Fig. 11. SPM results achieved by various algorithms.

TABLE I
STATISTICS OF EACH CLASS FOR HOUSTON DATASET

Class	*Number of labeled pixels*	*Number of training pixels*	*Training ratio (%)*
grass_healthy	*58553*	*42638*	*72.82*
grass_stressed	*111828*	*80369*	*71.87*
grass_synthetic	*983*	*983*	*100*
tree	*48757*	*39180*	*80.36*
soil	*40785*	*32110*	*78.73*
water	*420*	*420*	*100*
residential	*69479*	*51776*	*74.52*
commercial	*49559*	*34123*	*68.85*
road	*68195*	*45571*	*66.82*
highway	*53227*	*28407*	*53.37*
railway	*30902*	*23153*	*74.92*
parking_lot1	*37470*	*29059*	*77.55*
parking_lot2	*13094*	*10349*	*79.04*
tennis_court	*8600*	*6075*	*70.64*
running_track	*2057*	*1771*	*86.10*
total	***593909***	***425984***	***71.72***

TABLE II
STATISTICS OF ENDMEMBERS FOR HOUSTON DATASET

Class	*Number of endmembers*	*Ratio (%)*	*Number of selected pixels*
grass_healthy	*605*	*72.82*	*441*
grass_stressed	*923*	*71.87*	*663*
grass_synthetic	*42*	*100*	*42*
tree	*503*	*80.36*	*404*
soil	*493*	*78.73*	*388*
water	*10*	*100*	*10*
residential	*940*	*74.52*	*700*
commercial	*1469*	*68.85*	*1011*
road	*765*	*66.82*	*511*
highway	*1218*	*53.37*	*650*
railway	*213*	*74.92*	*160*
parking_lot1	*651*	*77.55*	*505*
parking_lot2	*102*	*79.04*	*81*
tennis_court	*21*	*70.64*	*15*
running_track	*40*	*86.10*	*34*

though most of them are not without their shortcomings. Both SRM$_{CNN}$ and SPMCNN-ESPCN fall short in their ability to restore and enhance spatial details and contextual information. Furthermore, they are hindered by the extensive variety and intricate distribution of land cover types, leading to an insufficient recovery of each land cover's spatial distribution. CASNet employs a densely connected structure to preserve hierarchical low-level features in the network's deep layers. However, its inability to handle the complexity of the land surface is evident from the results. SCNet and SIMNet incorporate LR semantic information to modulate the network features and recover spatial contextual information. However, the modulation methods are simplistic and constrained by the spatial resolution of LR information. Consequently, the reconstruction of specific land cover types is found to be inadequate, most notably for those exhibiting a linear distribution pattern. SFRNet-MLFF, in contrast, exhibits more pronounced benefits. The design of its network structure facilitates a more comprehensive and precise recovery of spatial details and distribution, rendering the edge portions of the land cover more fluid. Taking "roads" and "highways" as examples, the results garnered by SFRNet-MLFF are visually more distinct and continuous. They are clearly distinguished from other land cover types, with smoother and more defined boundary areas, thereby offering superior visual representation and enhanced SPM effects.

The quantitative evaluation comparison results of various SPM methods are detailed in Table III. It is worth noting that the $F1$ scores for "grass_synthetic," "water," "tennis_court," and "running_track" are not provided because all or the majority of the pixels for these four categories are allocated to the training data, and they themselves only constitute a small number of pixels in the data, rendering their quantitative evaluation unjustifiable. When compared to other methods, SFRNet-MLFF has significantly enhanced the $F1$ score for each category in the Houston dataset. Concurrently, the OA and $\kappa(\times 100)$ of the proposed SFRNet-MLFF reach 75.07% and 71.73, respectively, which are 36.47% and 39.53, 24.28% and 27.39, 23.91% and 27.04, 24.63% and 27.70,

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TABLE III
QUANTITATIVE ASSESSMENT OF EACH ALGORITHM ON THE HOUSTON DATASET

Class	SASPM	PSA	SRCNN-SPM	SRM_{CNN}	SPMCNN-ESPCN	CASNet	SCNet	SIMNet	SFRNet-MLFF
grass_healthy	0.4619	0.3118	0.4730	0.5065	0.4784	0.4967	0.6715	0.6721	**0.7365**
grass_stressed	0.3287	0.3048	0.3530	0.5650	0.5166	0.5373	0.6533	0.6460	**0.7206**
grass_synthetic	-	-	-	-	-	-	-	-	-
tree	0.4819	0.4684	0.4773	0.4943	0.4887	0.4870	0.6789	0.6685	**0.7473**
soil	0.3519	0.3397	0.4257	0.4301	0.4164	0.3956	0.5617	0.5819	**0.6240**
water	-	-	-	-	-	-	-	-	-
residential	0.4312	0.4226	0.4594	0.4840	0.4485	0.4548	0.6569	0.6652	**0.7312**
commercial	0.3822	0.3608	0.4052	0.6712	0.6522	0.6847	0.8055	0.8503	**0.8624**
road	0.3806	0.3447	0.4709	0.4926	0.5442	0.4899	0.7154	0.7363	**0.7749**
highway	0.2100	0.2092	0.2145	0.6308	0.6707	0.6497	0.8068	0.8238	**0.8522**
railway	0.1424	0.1092	0.1837	0.2086	0.2917	0.2592	0.5846	0.6306	**0.6825**
parking_lot1	0.2482	0.2002	0.2841	0.3678	0.3933	03764	0.6007	0.6700	**0.7081**
parking_lot2	0.2385	0.2620	0.2828	0.1661	0.1933	0.2123	0.3517	0.5844	**0.6269**
tennis_court	-	-	-	-	-	-	-	-	-
running_track	-	-	-	-	-	-	-	-	-
OA (%)	34.16	31.65	38.60	50.79	51.16	50.44	68.35	70.34	**75.07**
$\kappa(\times 100)$	27.86	24.69	32.20	44.34	44.69	44.03	64.15	66.44	**71.73**

TABLE IV
STATISTICS OF EACH CLASS FOR PAVIA UNIVERSITY DATASET

Class	*Number of labeled pixels*	*Number of training pixels*	*Training ratio (%)*
asphalt	*37878*	*22299*	*58.87*
meadows	*68514*	*41736*	*60.92*
gravel	*4909*	*2816*	*57.36*
trees	*21897*	*12390*	*56.58*
metal sheets	*2428*	*2349*	*96.75*
bare soil	*36536*	*23133*	*63.32*
bitumen	*3011*	*1390*	*46.16*
bricks	*18479*	*10124*	*54.79*
shadows	*10348*	*6643*	*64.20*
total	***204000***	***122880***	***60.24***

TABLE V
STATISTICS OF ENDMEMBERS FOR PAVIA UNIVERSITY DATASET

Class	*Number of endmembers*	*Ratio (%)*	*Number of selected pixels*
asphalt	*775*	*58.87*	*456*
meadows	*2153*	*60.92*	*1312*
gravel	*66*	*57.36*	*38*
trees	*316*	*56.58*	*179*
metal sheets	*75*	*96.75*	*73*
bare soil	*540*	*63.32*	*342*
bitumen	*58*	*46.16*	*27*
bricks	*310*	*54.79*	*170*
shadows	*81*	*64.20*	*52*

6.72% and 7.58, and 4.73% and 5.29 higher than those of SRCNN-SPM, SRM_{CNN}, SPMCNN-ESPCN, CASNet, SCNet, and SIMNet, respectively, indicating superior performance.

B. Experiment 2: Pavia University Dataset

The Pavia University dataset, collected by the Reflective Optics System Imaging Spectrometer (ROSIS) sensor, encompasses the urban region of Pavia University in Italy. It comprises 103 spectral bands, spanning from 0.43 to 0.86 μm, and features an image size of 610 $\times$ 340 pixels. The spatial resolution of the dataset is 1.3 m.

In the experiment, an image subset of 600 $\times$ 340 [Fig. 12(a)] was first extracted from the original data to facilitate data processing. It was then downsampled to produce an LR image [Fig. 12(b)] in line with the experimental scheme. The LR image, along with the HR GT classified into nine land cover types [Fig. 12(c)], was cropped into 45 pairs of LR image patches and GT patches. These patches measured 103 $\times$ 16 $\times$ 16 and 1 $\times$ 64 $\times$ 64. Out of these, 30 random patch pairs were chosen for network training, while the remaining 15 pairs were used for testing. Before being fed into the network, the GT patch underwent one-hot processing to get a shape of 9 $\times$ 64 $\times$ 64 and was subsequently downsampled to derive the LR classification map of 9 $\times$ 16 $\times$ 16. A selection of the cropped patch pairs, along with their corresponding original HR images, are displayed in Fig. 12(d). Table IV provides the count of labeled pixels in the data and the proportion ρ of these pixels that were utilized for training. For the UTM methods, endmembers [Fig. 13(b)] of the LR image were derived in accordance with the experimental scheme. Pure pixels from each category involved in the SU were selected based on the ratio ρ, as listed in Table V, for the extraction of the endmembers' spectra and the generation of abundance maps [Fig. 13(c)]. Four test LR image patches were chosen to demonstrate the predicted HRC results, as depicted in Fig. 14. It was observed that for images with fewer land cover categories, such as the Pavia University dataset, the HC results of LR images could somewhat represent surface details in areas with simple spatial distributions. However, the impact of mixed pixels remains significant. In comparison, the SPM results are more adept at representing the HR surface details, but the UTM methods are still constrained by the SU results. This leads to some land cover category identification errors, even for the SRCNN-SPM method, which visually appears to be the best among them. In addition, they are heavily influenced by noise. In the end-to-end DLSPMNets, we observe superior overall results. In a similar vein to the Houston dataset, both SRM_{CNN} and SPMCNN-ESPCN exhibit limitations in restoring the spatial distribution of various land cover types and in reconstructing spatial geometric information. While the densely connected structure of CASNet provides slightly more detailed results, they still fall short of expectations. The results of SCNet and SIMNet show significant improvement, but they lack adequate control over intricate details, resulting in irregularities at the edges of the land cover. Furthermore, as depicted in Fig. 14, SIMNet's

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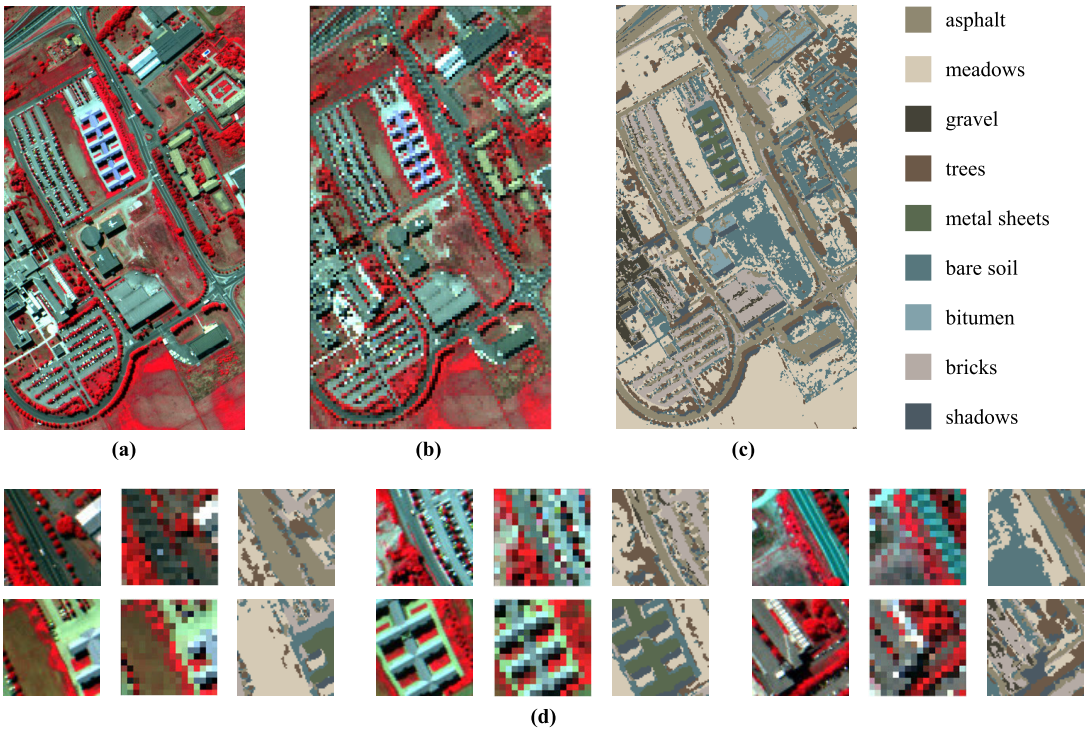


Fig. 12. Experiment on Pavia University dataset. (a) HR false-color image. (b) LR false-color image. (c) HR GT of the HR Pavia University image. (d) Part of the cropped obtained patch pairs and the corresponding original HR images.

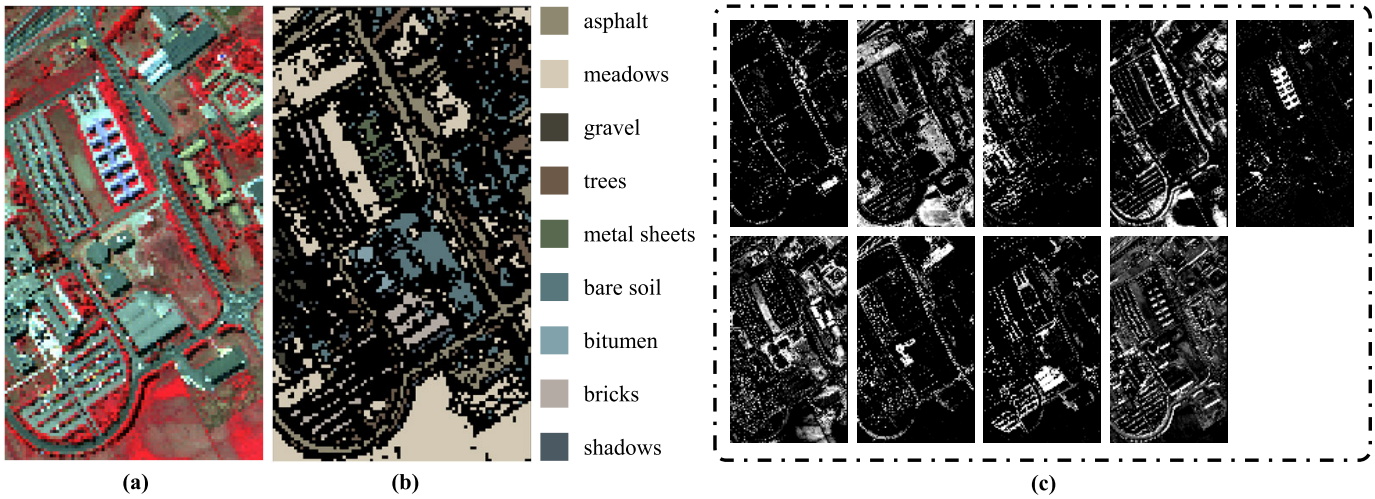


Fig. 13. Experiment on Pavia University dataset. (a) LR false-color image. (b) Pure pixels as endmembers. (c) Abundance maps of nine classes.

ability to differentiate between bitumen and asphalt could use further enhancement, and some land cover reconstruction results are still somewhat influenced by noise. SFRNet-MLFF, on the other hand, not only enhances the visual smoothness of the SPM results but also improves the ability to distinguish mixed land cover areas. This results in clear and regular land cover edges, minimizes noise interference, and produces results that align more closely with the corresponding GT patch.

The comparative results of the quantitative evaluation for various SPM methods are presented in Table VI. It is important to note that the majority of the pixels in the "metal sheets" were allocated to the training data, and they represent a small fraction of the data itself. This results in a lack of sufficient justification for the quantitative evaluation, and hence, no $F1$ score is provided for this category. When compared to other methods, SFRNet-MLFF demonstrates a substantial improvement in the $F1$ score for each category in the Pavia University dataset. Notably, the $F1$ score for the "bitumen" category has seen an increase of over 27% compared to the state-of-the-art DLSPMNets. Concurrently, the OA and $\kappa(\times 100)$ of our method reach 76.29% and 71.14 respectively, which are 29.58% and 33.87, 18.46% and 22.39, 24.03% and 29.87, 20.68% and 25.64, 10.98% and 13.70, and 3.54% and

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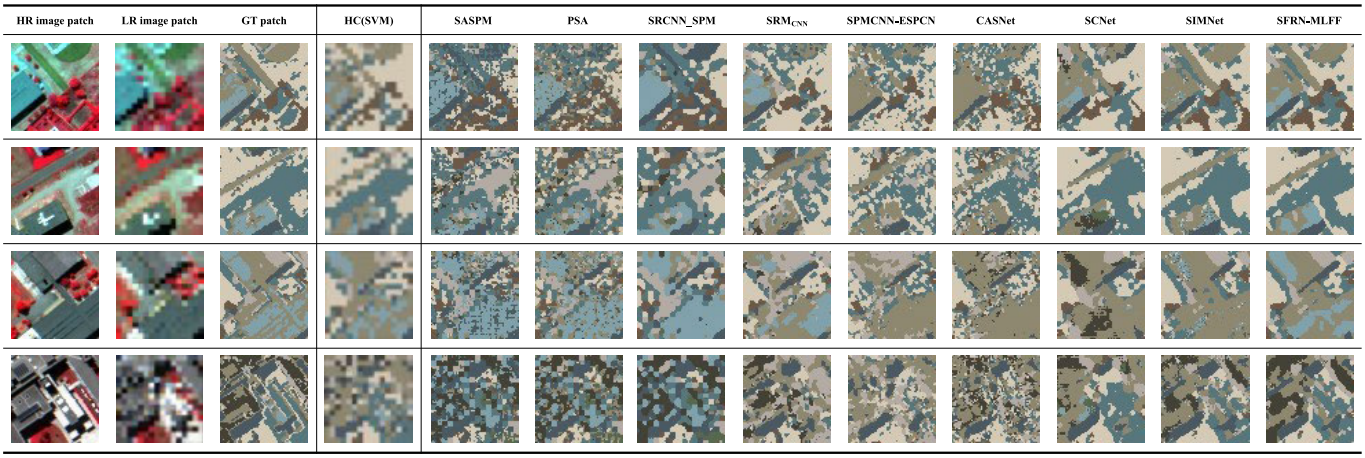


Fig. 14. SPM results achieved by various algorithms.

TABLE VI
QUANTITATIVE ASSESSMENT OF EACH ALGORITHM ON THE PAVIA UNIVERSITY DATASET

Class	SASPM	PSA	SRCNN-SPM	SRM$_{CNN}$	SPMCNN-ESPCN	CASNet	SCNet	SIMNet	SFRNet-MLFF
asphalt	0.2597	0.3473	0.2900	0.6003	0.5959	0.6233	0.6740	0.7565	**0.7905**
meadows	0.5226	0.5181	0.5777	0.6581	0.5982	0.6368	0.7517	0.7661	**0.7919**
gravel	0.3101	0.3225	0.3656	0.2281	0.1295	0.1572	0.2894	0.5964	**0.6525**
trees	0.5563	0.5615	0.5921	0.7083	0.6121	0.6262	0.6374	0.7394	**0.7700**
metal sheets	-	-	-	-	-	-	-	-	-
bare soil	0.3384	0.3198	0.3665	0.3907	0.3735	0.4032	0.6033	0.6666	**0.6928**
bitumen	0.2410	0.2581	0.2685	0.2806	0.2388	0.1738	0.1224	0.4224	**0.6982**
bricks	0.4415	0.4218	0.4752	0.5795	0.4493	0.5176	0.6799	0.7651	**0.7914**
shadows	0.3782	0.4820	0.4717	0.6427	0.5433	0.5977	0.5167	0.6545	**0.7316**
OA (%)	41.67	42.94	46.71	57.83	52.26	55.61	65.31	72.75	**76.29**
$\kappa(\times 100)$	32.21	32.19	37.27	48.75	41.27	45.50	57.44	66.68	**71.14**

4.46 higher than those of SRCNN-SPM, SRM$_{CNN}$, SPMCNN-ESPCN, CASNet, SCNet, and SIMNet, respectively, indicating a superior SPM performance.

C. Experiment 3: KSC Dataset

The KSC dataset was acquired by the National Aeronautics and Space Administration (NASA) Airborne Visible/Infrared Imaging Spectrometer (AVIRIS) instrument. The data were gathered over the KSC in Florida, USA, on March 23, 1996. After the removal of water absorption and low signal-to-noise ratio (SNR) bands from the original 224 bands, the dataset consists of 176 bands ranging from 0.4 to 2.5 μm. The image size is 512 $\times$ 614 pixels, with a spatial resolution of 18 m.

In the experiment, a subset of the image measuring 500 $\times$ 600 pixels was extracted from the original data for easier data processing, as shown in Fig. 15(a). This subset was then downsampled to create an LR image, depicted in Fig. 15(b), in accordance with the experimental design. The LR image and the HR GT, which was classified into 13 land cover types [Fig. 15(c)], were cropped into 63 pairs of LR image patches and GT patches with the shape of 176 $\times$ 16 $\times$ 16 and 1 $\times$ 64 $\times$ 64, respectively. After eliminating patch pairs with single or too few categories, 53 pairs remained. Out of these, 38 random patch pairs were chosen for network training, while the remaining 15 patch pairs were used for testing. The GT patch was processed using one-hot encoding to a shape of 13 $\times$ 64 $\times$ 64 before being input into the network. It was then downsampled to obtain the LR classification map with a size of 13 $\times$ 16 $\times$ 16. Fig. 15(d) displays a portion of the cropped patch pairs and their corresponding original HR images.

TABLE VII
STATISTICS OF EACH CLASS FOR KSC DATASET

Class	*Number of labeled pixels*	*Number of training pixels*	*Training ratio(%)*
water	*46155*	*28862*	*62.53*
mud flats	*20696*	*12158*	*58.75*
salt marsh	*17790*	*12594*	*70.79*
cattail marsh	*6337*	*4655*	*73.46*
spartina marsh	*29847*	*21457*	*71.89*
graminoid marsh	*14443*	*10417*	*72.12*
hardwood swamp	*2176*	*1799*	*82.67*
oak/broadleaf	*9179*	*7597*	*82.77*
slash pine	*263*	*225*	*85.55*
CP/oak	*9311*	*7035*	*75.56*
CP hammock	*8022*	*6019*	*75.03*
willow swamp	*6959*	*4820*	*69.26*
scrub	*45910*	*38010*	*82.79*
total	*217088*	*155648*	*71.70*

Table VII provides the number of labeled pixels and the proportion ρ of these pixels used for training. For the UTM methods, the endmembers [Fig. 16(b)] were derived from the LR image according to the experimental scheme. We then selected pixels from each category involved in the SU process. This allowed us to extract the spectra of the endmembers and generate the abundance maps [Fig. 16(c)], based on the ρ ratio (Table VIII). It is worth noting that the slash pine category

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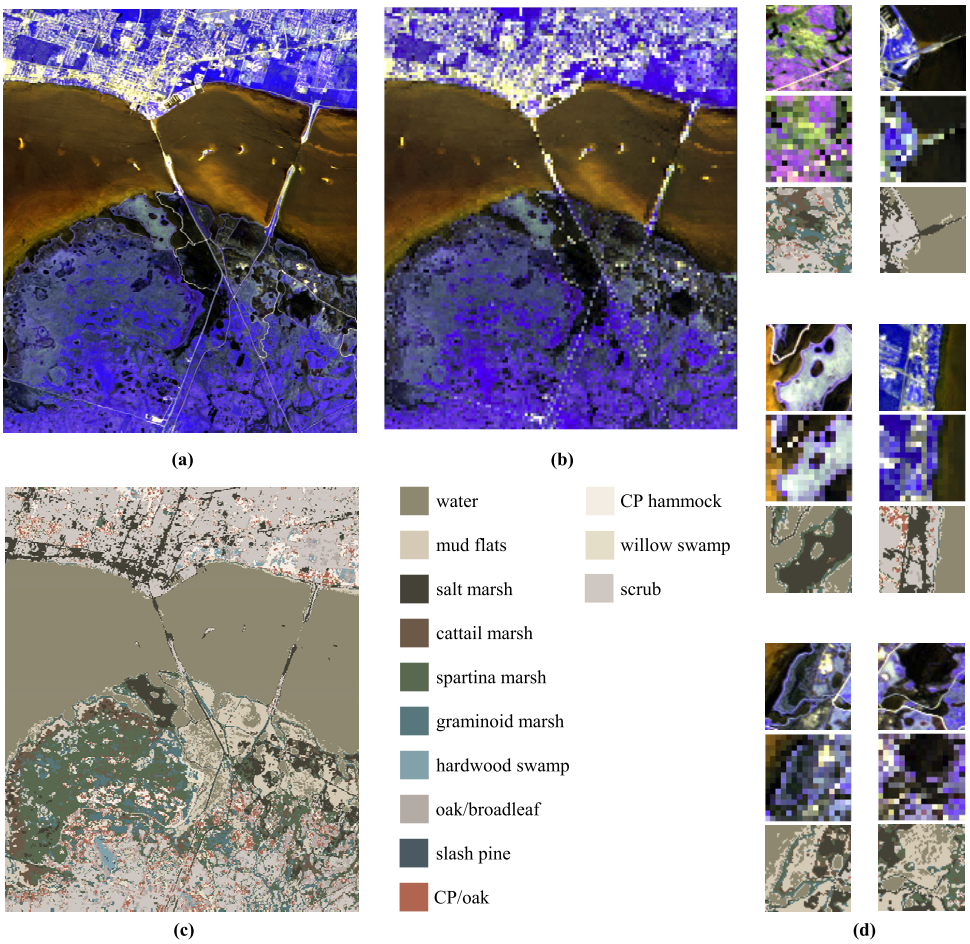


Fig. 15. Experiment on KSC dataset. (a) HR false-color image. (b) LR false-color image. (c) HR GT of the HR KSC image. (d) Part of the cropped obtained patch pairs and the corresponding original HR images.

TABLE VIII
STATISTICS OF ENDMEMBERS FOR KSC DATASET

Class	*Number of endmembers*	*Ratio (%)*	*Number of selected pixels*
water	*5536*	*62.53*	*3462*
mud flats	*225*	*58.75*	*132*
salt marsh	*217*	*70.79*	*154*
cattail marsh	*50*	*73.46*	*37*
spartina marsh	*465*	*71.89*	*334*
graminoid marsh	*50*	*72.12*	*36*
hardwood swamp	*17*	*82.67*	*14*
oak/broadleaf	*15*	*82.77*	*12*
slash pine	*0*	*85.55*	*0*
CP/oak	*4*	*75.56*	*3*
CP hammock	*28*	*75.03*	*21*
willow swamp	*12*	*69.26*	*8*
scrub	*947*	*82.79*	*784*

had too few pixels to select endmembers from, resulting in the absence of a corresponding abundance map.

We selected four test LR image patches to display the generated SPM results, as depicted in Fig. 17. The land cover distribution in the KSC dataset is highly fragmented and sporadic, containing numerous intricate details. This presents a great challenge for SPM. It is noticeable that the HC results lack many spatial details of the land cover due to the impact of mixed pixels. As expected, the UTM SPM methods are plagued by severe misclassification and noise interference, failing to establish a distinct and unambiguous boundary between water and land regions. Among the end-to-end DLSPMNets, the performance of SPMCNN and SPMCNN-ESPCN is subpar, resulting in poor SPM outcomes. The densely connected structure in the CASNet is evidently incapable of handling such complex surface details, leading to unsatisfactory final SPM results. The SPM results from SCNet and SIMNet have shown marked improvements over the previously mentioned methods. However, they still lack in detail, particularly in the reconstruction of the linearly distributed areas of the "spartina marsh" and "graminoid marsh." In contrast, SFRNet-MLFF demonstrates a superior SPM performance on the fragmented land cover distribution present in the KSC dataset. It is able to more accurately recover the original surface distribution pattern and visually aligns better with the reference GT patch.

The quantitative evaluation comparison results for various SPM methods are presented in Table IX. Notably, the categories "hardwood swamp" and "slash pine" are predominantly allocated to the training data and constitute a minimal portion of the data itself. The "slash pine" category, in particular, cannot be extracted during the SU process due to its limited pixel count. This renders the quantitative evaluation of these two categories somewhat unreasonable, and hence, their corresponding $F1$ scores are not provided. When compared to

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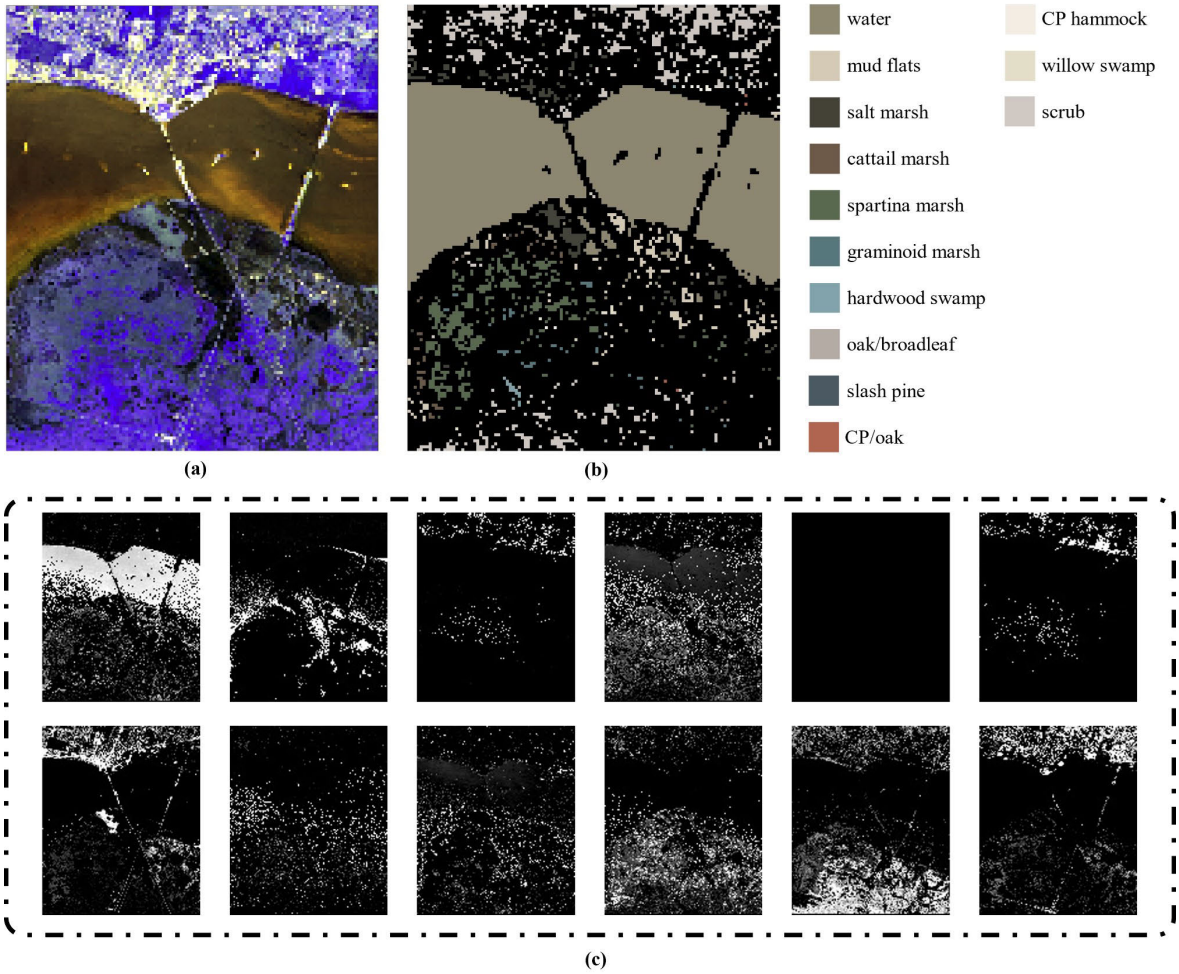


Fig. 16. Experiment on KSC dataset. (a) LR false-color image. (b) Pure pixels as endmembers. (c) Abundance maps of 12 available classes.

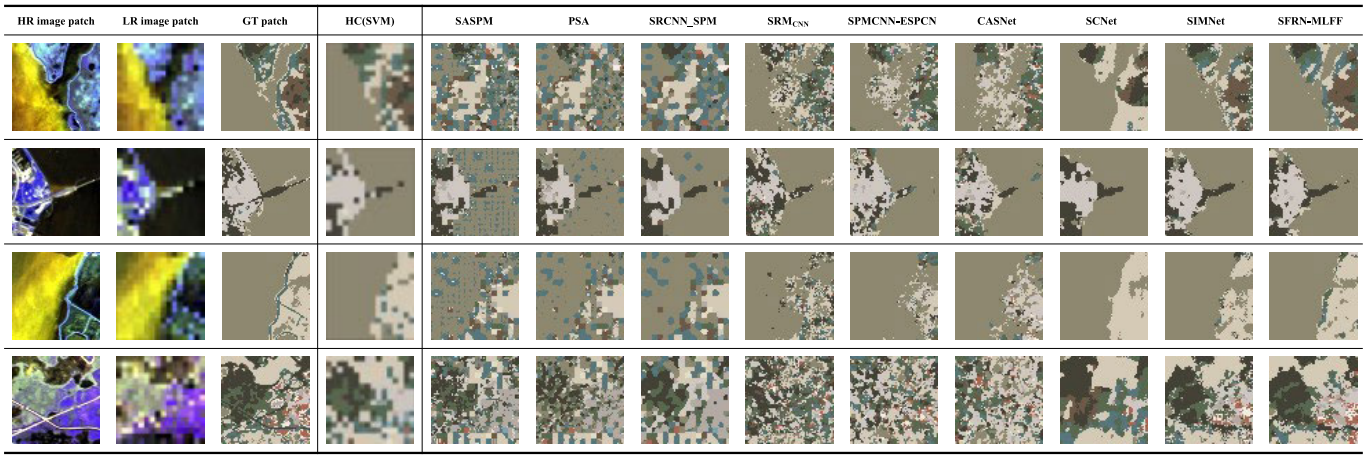


Fig. 17. SPM results achieved by various algorithms.

other SPM methods listed in the table, SFRNet-MLFF exhibits a significant improvement, with an increase of over 15% in the $F1$ score for the "cattail marsh" category, as well as substantial increases for other categories. Furthermore, the OA and $\kappa(\times 100)$ of our method reach 70.55% and 65.09, respectively, 26.43% and 30.43, 21.91% and 25.91, 23.4% and 27.96, 22.61% and 26.96, 7.17% and 8.77, and 4.31% and 5.28 higher than those of SRCNN-SPM, SRM$_{CNN}$, SPMCNN-ESPCN, CASNet, SCNet, and SIMNet, respectively, demonstrating superior SPM performance.

V. Methodological and Experimental Evaluation

A. Analysis of the Model

1) Feedback Architecture: To investigate the importance of utilizing varying resolution contexts in MRSFRA, we conducted additional comparative experiments. In these

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TABLE IX
QUANTITATIVE ASSESSMENT OF EACH ALGORITHM ON THE KSC DATASET

Class	SASPM	PSA	SRCNN-SPM	SRM_{CNN}	SPMCNN-ESPCN	CASNet	SCNet	SIMNet	SFRNet-MLFF
water	0.6686	0.6234	0.7402	0.8066	0.7843	0.7638	0.8917	0.9078	**0.9228**
mud flats	0.3224	0.3227	0.3359	0.3617	0.2518	0.3070	0.6264	0.6647	**0.7074**
salt marsh	0.4751	0.4684	0.5392	0.3928	0.3509	0.4021	0.5693	0.5804	**0.6467**
cattail marsh	0.0400	0.0378	0.0294	0.0727	0.0971	0.0669	0.1273	0.4196	**0.5724**
spartina marsh	0.2980	0.2565	0.3075	0.4393	0.4690	0.5119	0.5915	0.6232	**0.6694**
graminoid marsh	0.1583	0.1376	0.1955	0.1860	0.1676	0.1804	0.3040	0.3436	**0.4234**
hardwood swamp	-	-	-	-	-	-	-	-	-
oak/broadleaf	0.2308	0.2662	0.2390	0.1819	0.1664	0.1953	0.1911	0.3222	**0.4156**
slash pine	-	-	-	-	-	-	-	-	-
CP/oak	0.0770	0.0842	0.0902	0.1406	0.1475	0.1732	0.2150	0.2616	**0.3495**
CP hammock	0.3227	0.3139	0.3228	0.3497	0.2746	0.4495	0.5255	0.5291	**0.5929**
willow swamp	0.0283	0.0145	0.0110	0.0642	0.0876	0.0425	0.0538	0.2614	**0.3492**
scrub	0.4457	0.3965	0.4805	0.5100	0.5137	0.4676	0.6977	0.6706	**0.7307**
OA (%)	39.88	40.70	44.12	48.64	47.15	47.94	63.38	66.24	**70.55**
$\kappa(\times 100)$	30.10	27.45	34.66	39.18	37.13	38.13	56.32	59.81	**65.09**

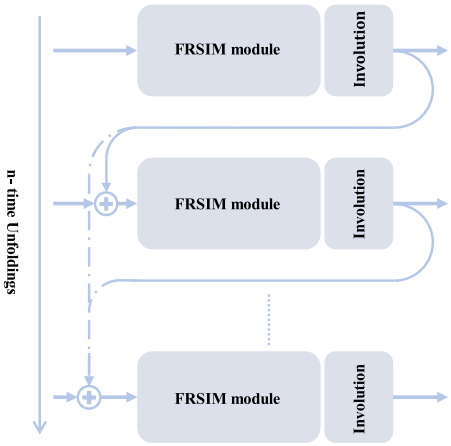


Fig. 18. Structure of FRSFRA.

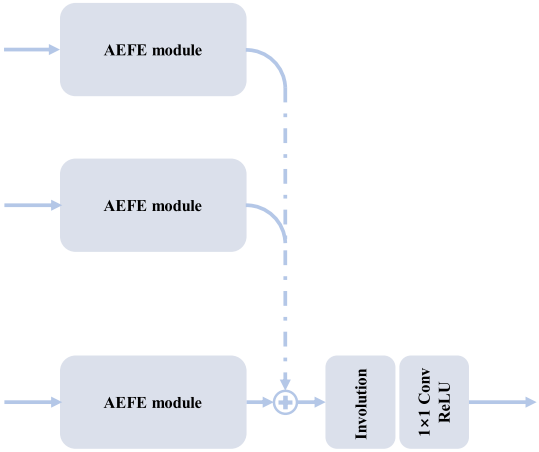


Fig. 19. PWA-based feature fusion.

TABLE X
QUANTITATIVE ASSESSMENT OF COMPARATIVE EXPERIMENTS ON THE FEEDBACK METHODS

Feedback Architecture	OA (%)		
	Houston	Pavia University	KSC
MRSFRA	**75.07**	**76.29**	**70.55**
FRSFRA	68.98	71.52	69.02

TABLE XI
QUANTITATIVE ASSESSMENT OF COMPARATIVE EXPERIMENTS ON THE FEATURE FUSION METHODS

Feature Fusion Method	OA(%)		
	Houston	Pavia University	KSC
CAFF	**75.07**	**76.29**	**70.55**
PWA	72.31	75.60	66.50

experiments, we maintained all other settings constant but fixed the spatial resolution of the network features to the LR context. This resulted in a fixed-resolution spatial feedback refined architecture (FRSFRA), as depicted in Fig. 18. Within the fixed-resolution spatial information modulation (FRSIM) module, the SPC was removed to ensure that the spatial resolution of the input and output features of the module remained consistent.

Table X presents the statistical results for the OA in the comparative experiments. The results indicate that modulating the features at a fixed resolution spatially leads to a substantial decline in the network's SPM performance when compared to the MRSFRA.

2) Feature Fusion Method: Feature fusion enables the aggregation of multilevel feature information. To illustrate the beneficial impact of CAFF on the SPM performance of the proposed network, we conducted comparative experiments focusing on feature fusion methods. In these experiments, all other settings remained constant, and PWA was employed as a comparative feature fusion method, as depicted in Fig. 19.

Table XI displays the OA from the comparative experiments. The statistical outcomes highlight a notable accuracy enhancement when using CAFF as opposed to PWA. This improvement evidently contributes to the effectiveness of SPM.

3) Unfolding Times n: To ascertain the ideal number of unfolding times (n) for SFRNet-MLFF, we carried out preliminary experiments using three different datasets. We tested a series of consecutive values for n while maintaining all other parameters constant. The OA and $\kappa(\times 100)$ corresponding to each n value were documented, as presented in Table XII. The results indicate that the OA and $\kappa(\times 100)$ peak at $n = 4$, with the performance slightly deteriorating before and after this point.

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TABLE XII
OA AND KAPPA ANALYSIS FOR DIFFERENT VALUES OF n

Datasets	Metrics	Value of Unfolding Times n						
		2	3	4	5	6	7	8
Houston	OA (%)	74.21	74.36	**75.02**	74.62	74.11	71.12	70.77
	$\kappa(\times 100)$	70.72	70.92	**71.64**	71.18	70.68	67.20	66.94
Pavia University	OA (%)	74.92	76.09	**76.29**	76.08	71.56	69.37	68.36
	$\kappa(\times 100)$	69.30	70.81	**71.14**	70.89	65.14	62.59	61.20
KSC	OA (%)	68.73	69.14	**70.55**	69.47	68.31	67.77	62.37
	$\kappa(\times 100)$	63.12	63.52	**65.09**	63.84	62.58	61.99	55.91

TABLE XIII
QUANTITATIVE ASSESSMENT OF ABLATION EXPERIMENTS

	Multi-Resolution Spatial Feedback Refined Architecture	Multi-Level Feature Fusion	Global Spatial Information Modulation Module	Houston	Pavia University	KSC
1	√	√	√	**75.07**	**76.29**	**70.55**
2	√	-	√	70.97	75.19	59.41
3	√	√	-	62.70	73.78	57.43
4	-	√	√	73.24	75.13	68.86
5	-	-	-	48.74	73.54	53.74

TABLE XIV
COMPUTATIONAL COST OF DIFFERENT DLSPMNETS

		SRM_{CNN}	SPMCNN-ESPCN	CASNet	SCNet	SIMNet	SFRNet-MLFF
Time(s)	Houston	8.670	6.498	10.540	3.092	10.875	10.848
	Pavia University	1.820	1.774	2.298	0.847	2.579	2.788
	KSC	2.997	2.582	2.9334	1.213	3.337	4.517
FLOPs	Houston	1.416G	1.054G	4.561G	4.688G	75.149G	57.853G
	Pavia University	1.361G	951.370M	4.458G	4.536G	73.870G	56.643G
	KSC	1.427G	1.046G	4.552G	4.744G	74.830G	57.554G
Params	Houston	520.111K	426.055K	4.175M	15.180M	65.797M	21.626M
	Pavia University	492.137K	397.675K	4.147M	15.047M	64.775M	21.248M
	KSC	540.013K	442.827K	4.192M	15.240M	65.561M	21,603M

B. Ablation Study

To evaluate the contribution of each component in SFRNet-MLFF, including MRSFRA, MLFF, and GSIM, we performed ablation studies while maintaining all other experimental conditions constant. Table XIII displays the OA statistics from these ablation studies. The results indicate that the removal of any of the three modules leads to a decrease in network performance to varying extents. In contrast, the original network exhibits a significant improvement in effectiveness, with increases of 26.33%, 2.75%, and 16.81% on the three datasets compared to a baseline network without any of the primary modules. This evidence underscores the crucial role each module plays in enhancing the SPM performance of the proposed network.

C. Computational Cost

Table XIV presents a comparison of various DLSPMNets, focusing on the average training time per epoch, the count of trainable parameters, and the floating-point operations (FLOPs). The findings reveal that the average training duration for SFRNet-MLFF is comparable to that of CASNet and SIMNet and remains within acceptable bounds when contrasted with other models. Concurrently, despite its intricate architecture, SFRNet-MLFF's FLOPs and the quantity of network parameters are significantly lower than those of SIMNet.

VI. CONCLUSION

In this study, we present an SFRNet-MLFF for HSI SPM to address the mixed pixel issue. SFRNet-MLFF innovatively enhances SPM performance with MRSFRA and MLFF, which allow for the thorough use of features across different network levels. MRSFRA's multiunfolding feedback mechanism refines lower level features using higher level ones, generating robust multilevel features. MLFF integrates both spatial and CA to learn feature weights and employs cross-attention for the fusion of features across multiple network levels. The MRSIM in MRSFRA, along with the GSIM in the network, supplements spatial information, and imposes solution space constraints on the continuous resolution context from LR to HR, further improving SPM performance. The experimental results on three publicly available hyperspectral datasets show that SFRNet-MLFF significantly outperforms existing methods in terms of $F1$ scores, OA, and kappa coefficient, demonstrating superior SPM performance. SFRNet-MLFF's strength lies in its ability to enable full interaction of network features and ensure thorough use of feature information at all levels while maintaining solution space constraints. However, as a supervised learning method, SFRNet-MLFF heavily depends on labeled hyperspectral data, which is often scarce. Therefore, future research will focus on incorporating semi-supervised [57], [58], [59] or unsupervised [60], [61]

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methods to address the dataset shortage and achieve broader applications.

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